量子力学期末复习

12.23

一. 基础知识 (一) sch Equ

1. $-\frac{\hbar^2}{2m}\nabla^2\psi + V\psi = i\hbar\partial_t\psi$ PDE. 线性. 非相. C.

2. $\int_{-\infty}^{\infty}\psi^*\psi\,dx = 1$

$$\frac{d}{dt}\int_{-\infty}^{\infty}\psi^*\psi\,dx = \int_{-\infty}^{\infty}\psi^*\partial_t\psi + \psi\partial_t\psi^* \overset{\text{sch Equ}}{=} \frac{i\hbar}{2m}(\psi^*\partial_x\psi - \psi\partial_x\psi^*)\Big|_{-\infty}^{\infty} = 0$$

不积分：流守恒. $\partial_t(\psi^*\psi)+\partial_x\left(\frac{i\hbar}{2m}(\psi^*\partial_x\psi-\psi\partial_x\psi^*)\right)=0$

推广至三维: $\partial_t\rho+\nabla\cdot\vec{J}=0$

3. $\langle x\rangle = \int_{-\infty}^{\infty}\psi^* x\psi\,dx$ (概率诠释)

4. $\frac{d\langle x\rangle}{dt} \triangleq \frac{\langle p\rangle}{m} \Rightarrow \langle p\rangle = \frac{d}{dt}\int_{-\infty}^{\infty}\psi^*x\psi\,dx = \int_{-\infty}^{\infty}\partial_t(\psi^*\psi)\,x\,dx \overset{2,\ \text{IBP}}{=} \int_{-\infty}^{\infty}\psi^*(-i\hbar\partial_x)\psi\,dx$

$\Rightarrow \hat{p} = -i\hbar\partial_x \Rightarrow$ ① $[x,p]=i\hbar$. ② $H=\frac{p^2}{2m}+V$. ③ $H\psi=i\hbar\partial_t\psi$.

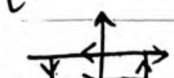
④ $\langle T\rangle = \int_{-\infty}^{\infty}\psi^*\left(\frac{-\hbar^2}{2m}\right)\partial_x^2\psi\,dx \overset{\text{IBP}}{=} \int_{-\infty}^{\infty}\frac{\hbar^2}{2m}|\partial_x\psi|^2dx \geq 0$, 用于积 $\partial_x\psi$ 不连续的.

· 以后 $\frac{d}{dt}\langle A\rangle$ 都是这样证, ∂_t, Sch Equ, IBP. $\frac{d}{dt}\langle A\rangle = \frac{i}{\hbar}[A,H] + \langle\partial_t A\rangle$

作业已证过: $\frac{d}{dt}\langle p\rangle = -\langle\partial_xV\rangle$, $\frac{d}{dt}\langle\vec{L}\rangle = \langle\vec{r}\times(-\nabla V)\rangle$, $\frac{d}{dt}\langle xp\rangle = 2\langle T\rangle - \langle x\partial_x V\rangle$

⑤ 算符 $\hat{p}e^{ikx} = -i\hbar\partial_x(e^{ikx}) = \hbar k\, e^{ikx} = p\, e^{ipx/\hbar}$ (记 $\overset{数}{p}=\hbar k$) $\hat{p}|p\rangle = p|p\rangle$

⑥
$$\begin{cases}\tilde{f}(k,t) = \int_{-\infty}^{\infty}dx\, f(x,t)\, e^{-ipx/\hbar} \to \int_{-\infty}^{\infty}\frac{dx}{\sqrt{2\pi\hbar}}f(x,t)e^{-ipx/\hbar} \\ f(x,t) = \int_{-\infty}^{\infty}\frac{dk}{2\pi}\tilde{f}(k,t)e^{ipx/\hbar} \to \int_{-\infty}^{\infty}\frac{dp}{\sqrt{2\pi\hbar}}\tilde{f}(p,t)e^{ipx/\hbar}\end{cases}$$
(对波函数)

tips: $\int_{-\infty}^{\infty}dp\, e^{-ipx/\hbar} = \lim_{\epsilon\to0}\int_{-\infty}^{\infty}dp\, e^{-\epsilon p^2}e^{-ipx/\hbar}$, 配方, 回路, ,

$\epsilon\to0$ 时 高斯函数 $\to\delta$ 函数 $\mapsto \sqrt{2\pi\hbar}\,\delta(p)$.

⑦ $\hat{p}$ 在动量空间的表示: $\langle p\rangle = \int_{-\infty}^{\infty}dx\,\psi^*(x,t)(-i\hbar\partial_x)\underbrace{\psi(x,t)}_{=\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\hbar}}dp\,\tilde{\psi}(p,t)e^{ipx/\hbar}}$ $-i\hbar\cdot\frac{ip}{\hbar}=p$ (数)

$$=\int_{-\infty}^{\infty}\frac{dp}{\sqrt{2\pi\hbar}}\left[\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\hbar}}dx\,\psi(x,t)e^{-ipx/\hbar}\right]^* p\,\tilde{\psi}(p,t)\cdot\sqrt{2\pi\hbar}$$

$\hat{p}=p$

$$=\int_{-\infty}^{\infty}dp\,\tilde{\psi}^*(p,t)\,p\,\tilde{\psi}(p,t).$$

$-i\hbar\partial_p(e^{-ipx/\hbar}) = x$

$\hat{x}$: $\langle x\rangle = \int_{-\infty}^{\infty} dx\, \psi^*(x,t)\, x\, \underbrace{\psi(x,t)}_{\int_{-\infty}^{\infty}\frac{dp}{\sqrt{2\pi\hbar}}\tilde{\psi}(p,t)e^{ipx/\hbar}}$

IBP

$= \int_{-\infty}^{\infty} dx\, \psi^*(x,t) \int_{-\infty}^{\infty} \frac{dp}{\sqrt{2\pi\hbar}} \tilde{\psi}(p,t)\cdot(-i\hbar\partial_p)e^{ipx/\hbar}$

$= \int_{-\infty}^{\infty} dp\, \left[\int_{-\infty}^{\infty}\frac{dx}{\sqrt{2\pi\hbar}}\psi(x,t)e^{-ipx/\hbar}\right]^* i\hbar\partial_p\tilde{\psi}(p,t)$

$= \int_{-\infty}^{\infty} dp\, \tilde{\psi}^*(p,t)\,(i\hbar\partial_p)\,\tilde{\psi}(p,t)$ $\qquad \hat{x} = i\hbar\partial_p$

Sch Equ: $\left[\frac{\hat{p}^2}{2m}+V(x)\right]\psi(x,t) = i\hbar\partial_t\psi(x,t)$. 左右 $\int_{-\infty}^{\infty}\frac{1}{\sqrt{2\pi\hbar}}dx\,e^{-ipx/\hbar}$

IBP. $V(x)=\sum_{n=0}^{\infty} f_n x^n$. $x^n \to (i\hbar\partial_p)^n$. 提出$\partial_p$. $\Rightarrow \frac{p^2}{2m}\tilde{\psi}(p,t)+V(\hat{x}=i\hbar\partial_p)\tilde{\psi}(p,t)$

或写为(更一般): $\frac{p^2}{2m}\tilde{\psi}(p,t)+\frac{1}{\sqrt{2\pi\hbar}}\int_{-\infty}^{\infty}\tilde{V}(p-p')\tilde{\psi}(p',t)dp' = i\hbar\partial_t\tilde{\psi}(p,t)$

积分方程. 需无穷多个边界条件(某p处$\tilde{\psi}(p)$取值依赖于所有其他p处$\tilde{\psi}$值

(类似的. $H|\psi\rangle = E|\psi\rangle \Rightarrow E\langle x|\psi\rangle = \langle x|H|\psi\rangle = \int \langle x|H|x'\rangle\langle x'|\psi\rangle dx'$

即 $E\psi(x) = \int_{-\infty}^{\infty}\langle x|H|x'\rangle\,\psi(x')dx' = (H\psi)(x)$. 一般的乘法算符如$V(x)$

意味着 $\langle x|V|x'\rangle = V(x)\delta(x-x')$. 即非对角元为0. $T=\frac{p^2}{2m}$ 在动量表象中

是乘法算符. $\langle p|T|p'\rangle = \frac{p^2}{2m}\delta(p-p')$. 故 $\langle x|T|x'\rangle = \iint_{-\infty}^{\infty}\langle x|p\rangle\langle p|T|p'\rangle\langle p'|x'\rangle dp\,dp'$

5. 解 sch Equ: 先解定态(H的本征方程). $\Psi(x,t)=\psi(x)f(t)$. 分离变量

$H\Psi(x,t) = i\hbar\partial_t\Psi(x,t) = i\hbar\psi(x)\partial_t f(t) = H\psi(x)f(t) \Rightarrow \frac{1}{\psi(x)}H\psi(x) = \frac{i\hbar\partial_t f(t)}{f(t)} \triangleq E$

($f(t) = Ae^{-iEt/\hbar} \Rightarrow \Psi(x,t) = \sum_n c_n\psi_n(x)e^{-iE_nt/\hbar}$), $H\psi_n(x) = E_n\psi_n(x)$.

· 或理解为: 用H的本征函数系来展开$\Psi(x,t) = \sum_n c_n(t)\psi_n(x)$, 代回 $H\Psi = i\hbar\partial_t\Psi$:

$H\sum_m c_m(t)\psi_m(x) = i\hbar\partial_t\sum_n c_n(t)\psi_n(x)$, 左 $\psi_i^*(x)$. $\int_{-\infty}^{\infty}\psi_i^*(x)\sum_m c_m(t)E_m\psi_m(x) = i\hbar\cdot$

$\int_{-\infty}^{\infty}dx\sum_n \dot{c}_n(t)\psi_i^*(x)\psi_n(x) \Rightarrow E_n c_n(t) = i\hbar\dot{c}_n(t) \Rightarrow c_n(t) = b_n e^{-iE_nt/\hbar}$.

$\langle\hat{O}\rangle = \int\Psi^*\hat{O}\Psi dx = \int dx\sum_m b_m^* e^{iE_mt/\hbar}\psi_m^*(x)\,\hat{O}\sum_n b_n e^{-iE_nt/\hbar}\psi_n(x)$

$= \sum_{mn} b_m^* b_n \langle m|\hat{O}|n\rangle e^{i(E_m-E_n)t/\hbar}$. 如果$[O,H]=0$, 即$O_{mn}$对角. 则不随$t$变.

则$= \sum_n |b_n|^2\lambda_n$ $\qquad \lambda_n$: 可能取值($\hat{O}$的本征值). $|b_n|^2$: 概率(叠加系数)

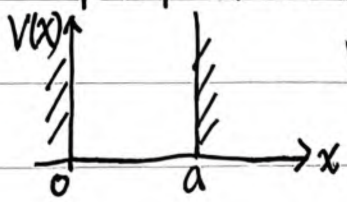
· 若不用H而用一般算符$\hat{O}$的本征函数展开$\Psi(x,t) = \sum_n c_n(t)\psi_n(x)$, 则 sch Equ $\Rightarrow$

$i\hbar\dot{c}_n(t) = \sum_m\langle n|H|m\rangle c_m(t)$, 如果$[H,O]\neq 0$ 则为矩阵方程.

· $\Psi(x,t)$ 不是$\hat{H}$的本征函数. 因为$\hat{H}\Psi = i\hbar\partial_t\Psi$, ∂_t是算符, 不是数.

(二) 模型(可精确求解的)

1. 一维无限深势阱



$V(x)=\begin{cases}\infty, & x\in[0,a] \\ 0, & \cdots\end{cases}$

方程:

$x<0$: $\psi(x)=0$, $x>a$: $\psi(x)=0$

$0\le x\le a$: $-\frac{\hbar^2}{2m}\frac{d^2}{dx^2}\psi(x)=E\psi(x)$

$\psi''+\frac{2mE}{\hbar^2}\psi(x)=0$. $k^2=\frac{2mE}{\hbar^2}$

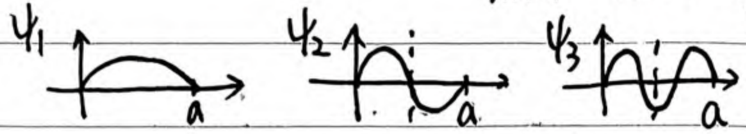
二重简并. $k, -k$. e^{ikx}, e^{-ikx}为2个基.

(或重新组合以$\sin kx, \cos kx$为2个基)

边界条件: $\psi(x)$连续

$$\begin{cases}\psi(0^-)=\psi(0^+) \Rightarrow \sim \sin kx \\ \psi(a^-)=\psi(a^+) \Rightarrow ka=n\pi \Rightarrow E_n=n^2\frac{\pi^2\hbar^2}{2ma^2}\end{cases}$$

$$\psi_n(x)=A_n\sin\left(\frac{n\pi}{a}x\right) \overset{\text{归一化}}{=} \sqrt{\frac{2}{a}}\sin\left(\frac{n\pi}{a}x\right).$$



$$\Psi(x,t)=\sum_n c_n\psi_n(x)e^{-iE_nt/\hbar}$$

$$c_n=\int_0^a\psi_n^*(x)\Psi(x,0)\,dx$$

① $V(x)$关于$\frac{a}{2}$对称, 故$\psi(x)$关于$\frac{a}{2}$是固定宇称的(奇或偶).

② $n\uparrow, E\uparrow$, 节点越多. 第n个有$n-1$个节点 (第$n-1$个激发态)

③ H的本征函数必正交归一 $\int_{-\infty}^{\infty}\psi_m^*(x)\psi_n(x)dx=\delta_{mn}$ ← H表象基矢的正交归一性 (离散)

④ 完备性: $\forall f(x)$有$f(x)=\sum_n c_n\psi_n(x)$. $\Rightarrow \int\psi_m^*(x)f(x)dx=\int\psi_m^*(x)\sum_n c_n\psi_n(x)dx=c_m$

$$f(x)=\sum_n\left(\int\psi_n^*(x')f(x')dx'\right)\psi_n(x) \Rightarrow \delta(x-x')=\sum_n\psi_n^*(x')\psi_n(x)$$

即 $\langle x|x'\rangle=\sum_n\langle x|\psi_n\rangle\langle\psi_n|x'\rangle$. $|\psi_n\rangle$完备.

坐标表象基矢的正交归一性 (连续).

$|x\rangle$不是平方可积的, 不$\in H$. 因此$\langle x|x'\rangle=\delta(x-x')$是我们选择的, 以使其完备性成立($\int|x\rangle\langle x|dx=1$) $\langle x|\psi\rangle\cdots$

⑤ 本征值(能量)离散 ← 边条定

以上5条是一维束缚态共同特征.

2. δ势.

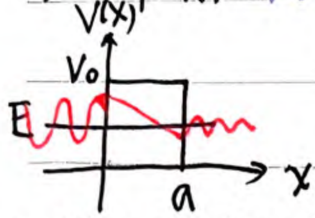
(1) δ势阱. $V(x)=-\alpha\delta(x)$. $\alpha>0$.

① 束缚态

② 散射态

(2) δ势垒 $V(x)=\alpha\delta(x)$

3. 一维有限高方势垒



$V(x)=\begin{cases} V_0, & 0<x<a \\ 0, & \cdots \end{cases}$ 左侧入射. (1) $E<V_0$ (隧穿)

$\psi''(x)+\frac{2m}{\hbar^2}(E-V)\psi(x)=0$

方程 → 得本征函数

① $x<0, x>a$: $V(x)=0$. $\psi''+\frac{2mE}{\hbar^2}\psi=0$. $\psi(x)\sim e^{\pm ikx}$. 左侧入射, 故

$$\psi(x)=\begin{cases} e^{ikx}+ce^{-ikx}, & x<0 \\ be^{ikx}, & x>a \end{cases} \qquad k=\frac{\sqrt{2mE}}{\hbar}$$

② $0<x<a$: $V(x)=V_0$. $\psi''-\frac{2m}{\hbar^2}(V_0-E)\psi=0$ $(i\beta)^2=-\beta^2$. $e^{\pm i(i\beta)x}=e^{\mp\beta x}$

$$\psi(x)=Ae^{\beta x}+Be^{-\beta x}, \quad 0<x<a. \qquad \beta=\frac{\sqrt{2m(V_0-E)}}{\hbar}$$ 或sh, ch

边界条件 → 定叠加系数.   ①+②, ①-②, ③+④, ③-④

在 $x=0$ ①②, $x=a$ ③④ 分别 $\psi(x)$ 连续、$\psi'(x)$ 连续 $\left(\begin{matrix} A=\cdots=\cdots \\ B=\cdots=\cdots \end{matrix}\right.$ 作比消掉)

$$\Rightarrow be^{ika}=\frac{-2ik\beta}{(\beta^2-k^2)\mathrm{sh}\beta a-2ik\beta\,\mathrm{ch}\beta a} \qquad \mathrm{ch}^2-\mathrm{sh}^2=1. \qquad \mathrm{sh}=\frac{e^{kx}-e^{-kx}}{2},\ \mathrm{ch}=\quad +$$

$$T=|b|^2=\left[1+\frac{(\beta^2+k^2)^2}{4k^2\beta^2}\mathrm{sh}^2\beta a\right]^{-1}=\left[1+\frac{V_0^2}{4E(V_0-E)}\mathrm{sh}^2\beta a\right]^{-1}$$

当 $\beta a\gg1$ (即 V_0 和/或 a 很大时. 很高和/或很宽): $\mathrm{sh}\beta a=\frac{e^{\beta a}-e^{-\beta a}}{2}\approx\frac{1}{2}e^{\beta a}\gg1$

$$T\approx\frac{4k^2\beta^2}{\beta^2+k^2}\cdot4e^{-2\beta a}=\frac{16k^2\beta^2}{\beta^2+k^2}e^{-2\beta a}\propto e^{-2\beta a}=\exp\left[-\frac{2a}{\hbar}\sqrt{2m(V_0-E)}\right]$$

WKB: $\psi''(x)+\underbrace{\frac{2m}{\hbar^2}(E-V_0)}_{f(x)}\psi(x)=0. \Rightarrow \psi(x)\approx f^{-\frac{1}{4}}(x)\exp\left(i\int dx\sqrt{f(x)}\right)$.

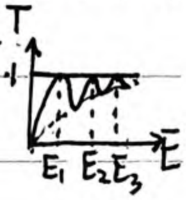
$$T\sim|\psi|^2\sim\exp\left(\frac{-2}{\hbar}\int dx\sqrt{2m(V_0-E)}\right)=\exp\left(\frac{-2a}{\hbar}\sqrt{2m(V_0-E)}\right).$$ 一致.

(2) $E>V_0$. 类似地分段求解. 中间段为 $\psi''+\frac{2m}{\hbar^2}(E-V_0)\psi=0$ (k_2^2). 是sin, cos (而非sh, ch)

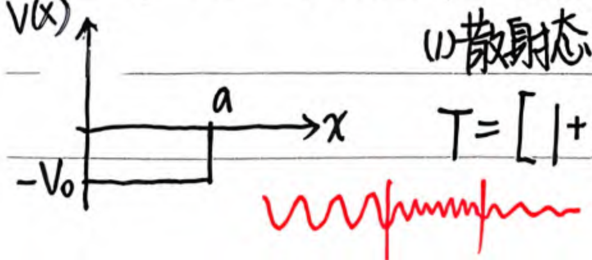
$$T=\left[1+\frac{V_0^2}{4E(E-V_0)}\sin^2k_2a\right]^{-1}$$ 当 $k_2a=n\pi$ 时, $T=1$. 共振透射.

$(\mathrm{sh}(ikx)=i\sin kx)$

$E_n=V_0+\frac{n^2\pi^2\hbar^2}{2ma^2}$



4. 一维有限深方势阱

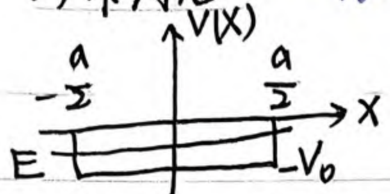


(1) 散射态. $E>0$. 则将3.(2)中 V_0 改为 $-V_0$ 即可.

$$T=\left[1+\frac{V_0^2}{4E(E+V_0)}\sin^2k_2a\right]^{-1}.$$ $k_2a=n\pi$ 时共振透射.

$$E_n=-V_0+\frac{n^2\pi^2\hbar^2}{2ma^2}$$

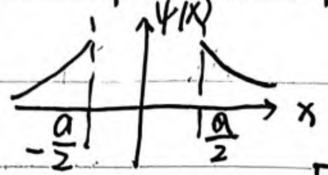
(2) 束缚态: $-V_0<E<0$



$$V(x)=\begin{cases}-V_0, & |x|<\frac{a}{2}\\ 0, & |x|>\frac{a}{2}\end{cases}$$

· 方程: ① $|x|>\frac{a}{2}$: $\psi''+\frac{2mE}{\hbar^2}\psi=0$. $E<0$. $\beta=\sqrt{\frac{-2mE}{\hbar^2}}$. $\psi(x)\sim e^{\pm\beta x}$. $|x|\to\infty$ 时 $\psi\to 0$:

$$\psi(x)=\begin{cases}Ae^{-\beta x}, & x\geq\frac{a}{2}\\ Be^{\beta x}, & x\leq-\frac{a}{2}\end{cases}$$



② $|x|\leq\frac{a}{2}$: $\psi''+\frac{2m(E+V_0)}{\hbar^2}\psi=0$. $E+V_0>0$. $k=\sqrt{\frac{2m(E+V_0)}{\hbar^2}}$. $\psi(x)\sim e^{\pm ikx}$.

$V(x)=V(-x)\Rightarrow\psi(x)$ 宇称固定. 为 $\sin kx$ 或 $\cos kx$.

· 边界条件: $\psi(x)$, $\psi'(x)$ 在 $x=\pm\frac{a}{2}$ 处连续. ($(\ln\psi)'$ 连续 ← 本征值)

(a.) 偶宇称态. $\psi(x)\sim\cos kx$.

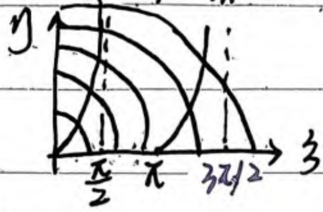
$$(\ln\psi)'=(\ln\cos kx)'\Big|_{x=\pm\frac{a}{2}}=(\ln e^{\mp\beta x})'\Big|_{x=\pm\frac{a}{2}}$$

$$k\tan(k\tfrac{a}{2})=\beta$$

$k\cdot\frac{a}{2}\tan(k\cdot\frac{a}{2})=\beta\cdot\frac{a}{2}$. $\xi=\frac{ka}{2}$, $\eta=\frac{\beta a}{2}$. $\xi\tan\xi=\eta$

$\xi^2+\eta^2=\frac{a^2}{4}\left(\frac{-2mE}{\hbar^2}+\frac{2m(E+V_0)}{\hbar^2}\right)=\frac{a^2}{4\hbar^2}2mV_0=\frac{ma^2V_0}{2\hbar^2}$ $\Rightarrow\begin{cases}\xi\tan\xi=\eta\\ \xi^2+\eta^2=\text{Const}=\frac{ma^2V_0}{2\hbar^2}\end{cases}$

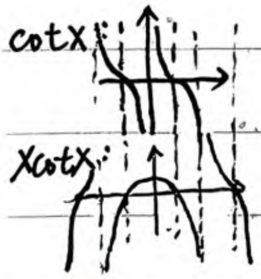
画出 $y=x\tan x$, $y^2+x^2=C$:
交点是方程组的解



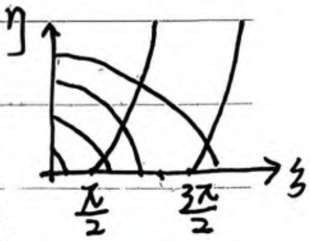
V_0 小解就少. 但至少有1个.
V_0 大解就多, 出现第二个条件: $\frac{ma^2V_0}{2\hbar^2}\geq\pi^2$

(b). 奇宇称态. $\psi(x)\sim\sin kx$

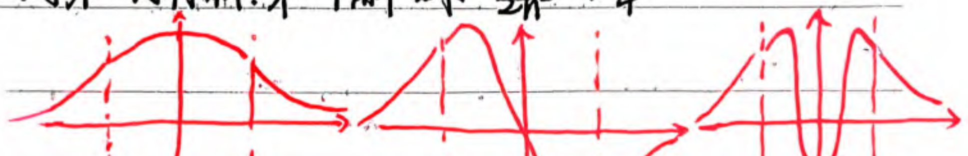
$$(\ln\psi)'=(\ln\sin kx)'\Big|_{x=\pm\frac{a}{2}}=(\ln e^{\mp\beta x})'\Big|_{x=\pm\frac{a}{2}}$$



$$-k\cot k\frac{a}{2}=\beta\Rightarrow\begin{cases}-\xi\cot\xi=\eta\\ \xi^2+\eta^2=\frac{ma^2V_0}{2\hbar^2}\end{cases}$$



V_0 很小时不一定有解. 第一个解要求: $\frac{ma^2V_0}{2\hbar^2}\geq\frac{\pi^2}{4}$



Q: 为什么束缚态的边界条件让本征值离散化, 散射态或势垒隧穿都不会, 边界条件只用来定系数? 核心: $x\to\infty$ 时 $\psi(x)\to 0$. 振荡要匹配, 光滑连接.

数学上，SL型问题在齐次边界条件下，本征值是离散的。束缚态要求 $\psi(x\to\pm\infty)=0$ 是齐次的。而散射态/势垒隧穿是给定了入射波，是非齐次边界条件。也无法齐次化。所以边条都用来定系数了。

· 为什么束缚态要求平方可积（$\psi(x\to\infty)=0$），而散射态不要求？因为 e^{ikx} 不是真实的物理态，是用来展开物理态的广义本征函数。不属于 $L^2(\mathbb{R})$。是散射问题的格林函数。

以上，通解很简单，难在用边条定本征值、系数

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5. 谐振子.（见 10.20 复习）
6. 氢原子.（见 11.8 复习）

} 有升降算符的方法.

7. 其他

(1) $V(x)=\dfrac{-V_0}{\cosh^2(ax)}$，$V_0>0$，$a>0$. 求能级.

$-\dfrac{\hbar^2}{2m}\psi''(x)-\dfrac{V_0}{\mathrm{ch}^2(ax)}\psi=E\psi$. $\boxed{\xi=\tanh(ax)}$ 则 $\dfrac{1}{\mathrm{ch}^2(ax)}=1-\xi^2$.

$\dfrac{d\xi}{dx}=(1-\xi^2)a$，$\dfrac{d\psi}{dx}=\dfrac{d\psi}{d\xi}\cdot\dfrac{d\xi}{dx}=(1-\xi^2)a\dfrac{d\psi}{d\xi}$，$\dfrac{d^2\psi}{dx^2}=\dfrac{d\xi}{dx}\cdot\dfrac{d}{d\xi}\left(\dfrac{d\psi}{dx}\right)=a^2(1-\xi^2)\dfrac{d}{d\xi}\left[(1-\xi^2)\dfrac{d\psi}{d\xi}\right]$

$\Rightarrow\ (1-\xi^2)\psi''-2\xi\psi'+\left(\gamma+\dfrac{\epsilon}{1-\xi^2}\right)\psi=0$. 连带勒让德. 若要在 $[-1,1]$ 上有限，则 $\gamma=\nu(\nu+1)$，$\epsilon=-\mu^2$，ν,μ 为整数. 解为连带勒让德多项式；若只需在 $(-1,1)$ 上有限（此处情况），则 $\nu-\mu=n$ 是整数即可. 解为连带勒让德函数

(2) 复习常见方程、边界条件、本征值、解（束缚态齐次边条）.

① $y''+\lambda y=0$.　$x\in[0,l]$，$y(0)=y(l)$　$\lambda_n=\left(\dfrac{n\pi}{l}\right)^2$，$e^{\pm ikx}$ 或 $\sin kx,\cos kx$

② $y''+\dfrac{1}{x}y'+\left(\lambda^2-\dfrac{\nu^2}{x^2}\right)y=0$.　$x\in[0,b]$，$y(0)$ 有界，$y(b)=0$.　$\lambda_n=\dfrac{\mu_n^{(\nu)}}{b}$.　$J_\nu\left(\dfrac{\mu_n^{(\nu)}}{b}x\right)$　第ν阶贝塞尔的第n个零点

③ $(1-x^2)y''-2xy'+\lambda y=0$.　$x\in[-1,1]$，$y(-1),y(1)$ 有界　$\lambda_l=l(l+1)$.　$P_l(x)$

④ $(1-x^2)y''-2xy'+\left(\lambda-\dfrac{m^2}{1-x^2}\right)y=0$.　$x\in[-1,1]$，$y(-1),y(1)$ 有界.　$\lambda_l=l(l+1)$　$P_l^m(x)$

⑤ $y''-2xy'+(\lambda-1)y=0$.　$x\in(-\infty,+\infty)$，$\lim\limits_{x\to\pm\infty}e^{-x^2/2}y=0$　$\lambda_n=2n+1$　$H_n(x)$

可变量替换换成它们. eg. $\phi''(x)+x\phi(x)=0$. 代换：$\phi=x^{\frac{1}{2}}y(t)$，$t=\frac{2}{3}x^{3/2}$

（常微分方程ODE. 没有待定的分离变量常数λ）

$\Rightarrow y''(t)+\frac{1}{t}y'(t)+\left(1-\dfrac{1}{9t^2}\right)y(t)=0.\ \Rightarrow J_{\frac{1}{3}},\ Y_{\frac{1}{3}}$（或 $J_{-\frac{1}{3}}$）

$\curvearrowleft [AB, C] \curvearrowright$

5. 谐振子

$H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2$ $\qquad H\psi_n(x) = (n+\frac{1}{2})\hbar\omega\,\psi_n(x)$

(1) 代数解法. $H = \frac{p^2}{2m} + \frac{1}{2}m\omega^2x^2 = (a^\dagger a + \frac{1}{2})\hbar\omega$. $\quad \hbar\omega\left(\sqrt{\frac{p^2}{2m\hbar\omega}} + \sqrt{\frac{m\omega}{2\hbar}x^2}\right)$

$$\begin{cases} a = \sqrt{\frac{m\omega}{2\hbar}}\left(x + \frac{ip}{m\omega}\right) \\ a^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(x - \frac{ip}{m\omega}\right) \end{cases} \Rightarrow \begin{cases} x = \sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a) \\ p = i\sqrt{\frac{\hbar m\omega}{2}}(a^\dagger - a) \end{cases}$$ 定义算符

有: $[a, a^\dagger] = 1$, $[H, a^\dagger] = \hbar\omega a^\dagger$, $[H, a] = -\hbar\omega a$ 求对易子

$$\Rightarrow \begin{cases} H(a^\dagger\psi) = (E+\hbar\omega)(a^\dagger\psi) \\ H(a\psi) = (E-\hbar\omega)(a\psi) \end{cases} \Rightarrow \begin{cases} a^\dagger|n\rangle = \sqrt{n+1}\,|n+1\rangle \\ a|n\rangle = \sqrt{n}\,|n-1\rangle \end{cases}$$ 证明升降 求系数

$\Rightarrow$ 基态波函数: $a\psi_0 = 0$ $\qquad \langle n|a\,a^\dagger|n\rangle \quad c^2$

$$x\psi + \frac{i}{m\omega}\cdot\left(-i\hbar\frac{d}{dx}\right)\psi = 0$$ 求波函数

$$\frac{d\psi}{dx} + \frac{m\omega}{\hbar}x\psi = 0$$

$$\psi_0 = Ae^{-\frac{m\omega}{2\hbar}x^2} = \left(\frac{m\omega}{\pi\hbar}\right)^{\frac{1}{4}} e^{-\frac{m\omega}{2\hbar}x^2}$$

$\Rightarrow$ 各级波函数: $\psi_n = \frac{1}{\sqrt{n!}}(a^\dagger)^n\psi_0(x)$

无量纲化: $\sqrt{\frac{m\omega}{\hbar}}x = \frac{x}{x_0} = \xi$. 则 $\psi_0 = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}e^{-\frac{1}{2}\xi^2}$, $a^\dagger = \frac{1}{\sqrt{2}}\left(\xi - \frac{d}{d\xi}\right)$

$$\Rightarrow \psi_n(x) = \frac{1}{\sqrt{2^n n!}}\left(\xi - \frac{d}{d\xi}\right)^n \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\frac{1}{2}\xi^2}$$

$$= \frac{1}{\sqrt{2^n n!}}\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} H_n(\xi)\, e^{-\frac{1}{2}\xi^2}$$

系数 $\qquad$ 厄米 × 高斯

$H_n(-x) = (-1)^n H_n(x)$ 奇/偶.

$H_0(\xi) = 1$.
$H_1(\xi) = 2\xi$
$H_2(\xi) = 4\xi^2 - 2$
$H_3(\xi) = 8\xi^3 - 12\xi$

应用: 求任何算符的任意矩阵元:

$O_{mn} = \langle m|\hat{O}(\hat{x}, \hat{p})|n\rangle$, 其中 $\hat{x} = \sqrt{\frac{\hbar}{2m\omega}}(a^\dagger + a)$. $\hat{p} = i\sqrt{\frac{\hbar m\omega}{2}}(a^\dagger - a)$

eg. $\langle x^2\rangle = \langle n|x^2|n\rangle = \frac{\hbar}{2m\omega}\langle n|\cancel{a^\dagger a^\dagger} + a^\dagger a + aa^\dagger + \cancel{aa}|n\rangle = (n+\frac{1}{2})\frac{\hbar}{m\omega}$

$\langle p^2\rangle = \langle n|p^2|n\rangle = -\frac{\hbar m\omega}{2}\langle n|\cancel{a^\dagger a^\dagger} - a^\dagger a - aa^\dagger + \cancel{aa}|n\rangle = (n+\frac{1}{2})\hbar m\omega$

$(\Delta x)^2 = \langle x^2\rangle - \langle x\rangle^2 \Rightarrow \sqrt{\Delta x\,\Delta p} = (n+\frac{1}{2})\hbar \geq \frac{1}{2}\hbar$ ✓

如果太高次幂, 积分算矩阵元即可. $\underbrace{\Gamma\text{函数}}$ ...

$$\int_0^\infty t^{p-1}e^{-t}dt = \Gamma(p). \quad \Gamma(\tfrac{1}{2}) = \sqrt{\pi}.$$

(2) 解方程(已知特殊方程和函数)

$$\psi'' + \frac{2m}{\hbar^2}\left(E - \frac{1}{2} m\omega^2 x^2\right)\psi = 0$$

设 $\psi_n(x) = y_n(x)\cdot e^{-\frac{m\omega}{2\hbar}x^2} = y_n(\xi)\, e^{-\frac{1}{2}\xi^2}$ 无量纲化 $\quad \frac{d}{dx} = \frac{1}{x_0}\frac{d}{d\xi}$

代回方程: $y''(\xi) - 2\xi y'(\xi) + \left(\frac{2mE x_0^2}{\hbar^2} - 1\right) y(\xi) = 0$

边界条件: $\lim\limits_{\xi\to\infty} y\, e^{-\frac{1}{2}\xi^2} = 0$

$\Rightarrow$ 厄米方程. $\lambda_n = \frac{2mE}{\hbar^2}\cdot\frac{\hbar}{m\omega} = 2n+1$. $\quad E_n = (n+\frac{1}{2})\hbar\omega$.

$y_n(\xi) = H_n(\xi)$.

如果不知道这个特殊方程和函数?

(3) 级数解法

$$\psi'' + \frac{2m}{\hbar^2}(E-V)\psi = 0$$

同样的, 设 $\psi_n(x) = y_n(\xi)\, e^{-\frac{1}{2}\xi^2}$. $\quad \frac{d}{dx} = \frac{1}{x_0}\cdot\frac{d}{d\xi}$.

代回方程: $y'' - 2\xi y' + \left(\frac{2E}{\hbar\omega} - 1\right) y = 0$ $\quad$ 边条: $\lim\limits_{\xi\to\infty} y\, e^{-\frac{1}{2}\xi^2} = 0$

设 $y_{\bar n}(\xi) = \sum\limits_n a_n \xi^n$. ξ^n 系数$=0$: $(n+2)(n+1)\,a_{n+2} = \left(2n+1 - \frac{2E}{\hbar\omega}\right) a_n$

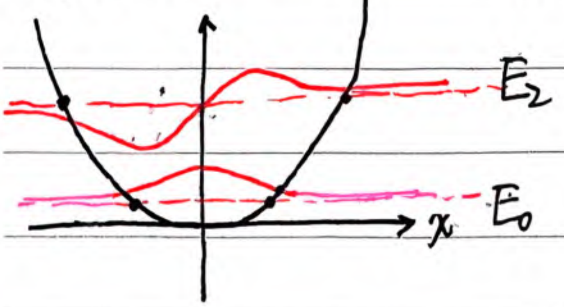
$$\frac{a_{n+2}}{a_n} = \frac{1}{(n+2)(n+1)}\left(2n+1 - \frac{2E}{\hbar\omega}\right) a_n.$$

若一直下去, $\frac{a_{n+2}}{a_n} \to \frac{2}{n}$. 则 $y(\xi) \to e^{\xi^2}$. $y(\xi) e^{-\frac{1}{2}\xi^2} \to e^{\frac{1}{2}\xi^2} \to \infty$. ✗

故要截断. 即让某个 $a_{\bar n+2} = 0$. $\Rightarrow E_{\bar n} = (\bar n + \frac{1}{2})\hbar\omega$.

$$y_{\bar n}(\xi) = \sum_{n=1}^{\bar n} a_n \xi^n.$$

(4) 图像 $\quad x_0 = \sqrt{\frac{\hbar}{m\omega}}$ 是基态禁区范围. $\frac{1}{2} m\omega^2 A^2 = \frac{1}{2}\hbar\omega$.



QM小测2. H原子. 复习 11.8

1. 方程 $\hat{H}\Psi = i\hbar\partial_t\Psi$ $\qquad V(r) = \frac{e^2}{4\pi\epsilon_0 r}$

↓定态

$$\left(-\frac{\hbar^2}{2m}\nabla^2 - \frac{e^2}{4\pi\epsilon_0 r}\right)\psi(\vec{r}) = E\psi(\vec{r})$$

↓球坐标 $\boxed{\nabla^2 = \frac{1}{r^2}\partial_r(r^2\partial_r) + \frac{1}{r^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta) + \frac{1}{r^2\sin^2\theta}\partial_\phi^2}$

$$\text{LHS} = \underbrace{-\frac{\hbar^2}{2mr^2}\partial_r(r^2\partial_r\psi)}_{\text{径向动能}}\underbrace{-\frac{\hbar^2}{2mr^2\sin\theta}\partial_\theta(\sin\theta\partial_\theta\psi) - \frac{\hbar^2}{2mr^2\sin^2\theta}\partial_\phi^2\psi}_{\text{角向动能}} - \frac{e^2}{4\pi\epsilon_0 r}\psi$$

一般 $= \frac{L^2}{2I} = \frac{L^2}{2mr^2}$

发现确实 $\boxed{\hat{L}^2 = (\vec{r}\times\vec{p})^2 = \frac{-\hbar^2}{\sin\theta}\left[\partial_\theta(\sin\theta\partial_\theta) + \frac{1}{\sin\theta}\partial_\phi^2\right]}$

(因为 $\nabla = \hat{r}\partial_r + \frac{1}{r}\partial_\theta\hat{\theta} + \frac{1}{r\sin\theta}\partial_\phi\hat{\phi}$. 代入即得)

$$E\psi(\vec{r}) = -\frac{\hbar^2}{2mr^2}\partial_r(r^2\partial_r\psi(\vec{r})) + \frac{\hat{L}^2}{2mr^2}\psi(\vec{r}) + V(r)\psi(\vec{r})$$

2. 解方程：分离变量 $\psi(r,\theta,\phi) = R(r)Y(\theta,\phi)$.

$$ER(r)Y(\theta,\phi) = -\frac{\hbar^2}{2mr^2}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right)\cdot Y(\theta,\phi) + R(r)\cdot\frac{\hat{L}^2}{2mr^2}Y(\theta,\phi) + VRY.$$

$$E = -\frac{\hbar^2}{2mr^2R}\frac{d}{dr}\left(r^2\frac{d}{dr}R\right) + \frac{\hat{L}^2Y}{2mr^2Y} + V(r)$$

$$[E-V(r)]\cdot 2mr^2 + \frac{\hbar^2}{R}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right) = \frac{\hat{L}^2Y(\theta,\phi)}{Y(\theta,\phi)} \triangleq C.$$

(1) 角向部分： $\hat{L}^2Y(\theta,\phi) = CY(\theta,\phi)$ $\qquad \hat{L}^2$的本征方程.

再分离变量： $Y(\theta,\phi) = \Theta(\theta)\Phi(\phi)$.

$$\frac{-\hbar^2}{\sin\theta}\left[\partial_\theta(\sin\theta\partial_\theta\Theta(\theta))\cdot\Phi(\phi) + \frac{1}{\sin\theta}\partial_\phi^2\Phi(\phi)\cdot\Theta(\theta)\right] = C\,\Theta(\theta)\Phi(\phi)$$

$$-\frac{1}{\Phi}\partial_\phi^2\Phi = \sin\theta\partial_\theta(\sin\theta\partial_\theta\Theta)\frac{1}{\Theta} + \frac{C}{\hbar^2}\sin^2\theta \triangleq K.$$

1° $\Phi(\phi)$部分： $\begin{cases}\partial_\phi^2\Phi = -K\Phi \\ \Phi(\phi) = \Phi(\phi+2\pi)\end{cases}$

通解：$\Phi(\phi)=Ae^{i\sqrt{k}\phi}+Be^{-i\sqrt{k}\phi}$ （线性独立，无固定字称）

考虑边条：$e^{i\sqrt{k}(\phi+2\pi)}=e^{i\sqrt{k}\phi}\cdot e^{i\sqrt{k}\cdot 2\pi} \Rightarrow e^{i\sqrt{k}\cdot 2\pi}=1$

$$\sqrt{k}=m,\quad m=0,1,2,\cdots$$

$\Rightarrow \Phi_m(\phi)\sim e^{im\phi}$，$m\in\mathbb{Z}$ (正负均可)

$$\Phi(\phi)=\sum_m A_m e^{im\phi}$$

△代回方程：$\partial_\phi^2\Phi_m=-m^2\Phi_m$. 当$m\neq0$时，一个本征值$(-m^2)$对应两个态.$(e^{im\phi}, e^{-im\phi})$

△ $\boxed{\hat{L}_z}=\vec{e}_z\cdot\hat{L}=(\hat{r}\cos\theta-\hat{\theta}\sin\theta)\frac{\hbar}{i}(\hat{\phi}\partial_\theta-\hat{\theta}\frac{1}{\sin\theta}\partial_\phi)\boxed{=-i\hbar\partial_\phi}$

$$\hat{L}_z\Phi_m(\phi)=-i\hbar\partial_\phi(e^{im\phi})=m\hbar\Phi_m.$$

$$\hat{L}_z^2\Phi_m(\phi)=m^2\hbar^2\Phi_m.$$

因此$\Phi_m(\phi)=e^{im\phi}$ $(m\in\mathbb{Z})$ 既作为$Y(\theta,\phi)$的组成部分，是$\hat{L}^2$的本征函数，其自身也是$\hat{L}_z=-i\hbar\partial_\phi$的本征函数.

2° $\Theta(\theta)$部分.

(a) 解析解法

$$\sin\theta\frac{d}{d\theta}(\sin\theta\frac{d}{d\theta}\Theta(\theta))\frac{1}{\Theta}+\frac{C}{\hbar^2}\sin^2\theta=k\underset{\text{由}1^\circ}{=}m^2$$

$$\sin\theta\frac{d}{d\theta}(\sin\theta\frac{d}{d\theta}\Theta(\theta))+[\frac{C}{\hbar^2}\sin^2\theta-m^2]\Theta=0$$

这是连带勒让德方程. 为了方程有解必须令$\frac{C}{\hbar^2}=l(l+1)$, $l\in\mathbb{Z}$.
这种离散化，让级数解能截断、收敛.

则解为 $\Theta(\theta)\sim P_l^m(\cos\theta)$ $\qquad P_l^m(x)$: $x\in[-1,1]$, $P_l(x)=(-1)^l P_l(-x)$

$$\Rightarrow\begin{cases} Y_l^m(\theta,\phi)\sim P_l^m(\cos\theta)e^{im\phi} \\ C=l(l+1)\hbar^2, & l\in\mathbb{Z} \\ k=m^2, & m\in\mathbb{Z}\end{cases}$$

角向部分没什么特别的，球对称情况下解都是这个.
因为能体现这个系统的$V(r)$只与r有关.

(b) 代数解法

已知：
$$\begin{cases} \hat{L}^2 Y(\theta,\phi) = C\,Y(\theta,\phi) \\ Y(\theta,\phi) = \Theta(\theta)\Phi(\phi) \\ \Phi_m(\phi) \sim e^{im\phi} \\ \hat{L}_z Y^m = -i\hbar\partial_\phi[\Theta(\theta)\Phi_m(\phi)] = \Theta(\theta)\cdot m\hbar\Phi_m(\phi) = m\hbar Y^m \end{cases}$$

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$$

$\hat{L}^2$ ↑ 想知作用后的本征值　　　$\hat{L}_z^2$ ↑ 已知作用后的本征值

$$= \underbrace{(L_x + iL_y)}_{L_+}\underbrace{(L_x - iL_y)}_{L_-} + i\underbrace{[L_x, L_y]}_{i\hbar L_z} + L_z^2$$

(def:)

$$\boxed{\hat{L}^2 = L_+L_- - \hbar L_z + L_z^2} = L_-L_+ + \hbar L_z + L_z^2$$

△ $L_\pm$ 的性质：

① $\boxed{L^2(L_\pm Y) = C(L_\pm Y)}$ 　不改变模量

证：$L^2(L_\pm Y) = (L_\pm L^2 + \underbrace{[L^2, L_\pm]}_{=0})Y = L_\pm L^2 Y \overset{L^2Y=CY}{=} L_\pm CY = C(L_\pm Y)$

② $\boxed{L_z(L_\pm Y) = (m\pm1)\hbar(L_\pm Y)}$ 改变z分量

证：$L_z(L_\pm Y) = (L_\pm L_z + \underbrace{[L_z, L_\pm]})Y = (L_\pm L_z \pm \hbar L_\pm)Y \overset{L_zY=m\hbar Y}{=} (m\pm1)\hbar(L_\pm Y)$.

$$= [L_z, L_x \pm iL_y] = i\hbar L_y \pm i(-i\hbar L_x) = \pm\hbar L_\pm$$

△ 使用 $L_\pm$：

① 说明 $C = l(l+1)\hbar^2$, $l \in \mathbb{Z}$：

由于作用 $L_\pm$ 只改分量不改模量，而分量不能比模量大，故有上下界.

即 $L_+Y^{m_{max}} = 0$，$L_-Y^{m_{min}} = 0$. 且 $m_{max} = -m_{min}$

eg. $L_+Y^{m_{max}} = 0$，则 $L^2Y^{m_{max}} = (L_-L_+ + \hbar L_z + L_z^2)Y^{m_{max}} = m_{max}(m_{max}+1)\hbar^2 Y^{m_{max}}$

故 $L^2Y = L^2Y^{m_{max}} = l(l+1)\hbar^2 Y$，$l = m_{max}$.

或者说，$m = -l, -l+1, \cdots, 0, \cdots, l-1, l$.

② 用 $L_+Y_l^l=0$ 得 Y_l^l，则 $Y_l^m = A(L_-)^{l-m}Y_l^l$

求 Y_l^l：$L_+Y_l^l=0$.

$$\boxed{L_+ = L_x + iL_y = \frac{\hbar}{i}e^{i\phi}(i\partial_\theta - \cot\theta\,\partial_\phi)}$$

$$Y_l^l = \Theta(\theta)e^{il\phi}$$

$$\Rightarrow (\cancel{i}\partial_\theta\Theta(\theta) - \cot\theta\cdot\Theta\cdot\cancel{i}l) = 0$$

$$\frac{d\Theta}{\Theta} = l\frac{\cos\theta\, d\theta}{\sin\theta} = l\frac{d(\sin\theta)}{\sin\theta}$$

$$\Theta(\theta) \sim \sin^l\theta$$

$$\Rightarrow \boxed{Y_l^l = A\sin^l\theta\, e^{il\phi}} \qquad Y_l^{-l} = A\sin^l\theta\, e^{-il\phi}$$

$$\boxed{L_- = L_x - iL_y = \frac{\hbar}{i}e^{i\phi}(-i\partial_\theta - \cot\theta\,\partial_\phi)}$$

不停作用，再归一化即可（定 $l \to m=-l,\cdots,l \to \cdots$）

$Y_0^0 = \frac{1}{\sqrt{4\pi}}$.

③ 升降时出的系数 $\begin{cases} L_-Y_l^m = cY_l^{m-1}, & c=? \\ L_+Y_l^m = c'Y_l^{m+1}, & c'=? \end{cases}$

使用 Y_l^m 的归一性.

两边同乘 $(L_\pm Y_l^m)^* = (cY_l^{m\pm1})^*$

eg. $L_-Y_l^m = cY_l^{m-1}$

$$\int d\Omega\, (L_-Y_l^m)^* L_-Y_l^m = \int d\Omega\, (L_-Y_l^m)^* cY_l^{m-1}$$

$$\int d\Omega\, Y_l^{m*}\underbrace{L_+L_-}_{L^2-L_z^2+\hbar L_z}Y_l^m = \int d\Omega\, (cY_l^{m-1})^* cY_l^{m-1} = |c|^2$$

$$= \int d\Omega\, [l(l+1)\hbar^2 - m^2\hbar^2 + m\hbar^2]\, Y_l^{m*}Y_l^m$$

$$\Rightarrow |c| = \sqrt{l(l+1)-m(m-1)}\,\hbar \qquad (L_-)$$

同理得 $|c'| = \sqrt{l(l+1)-m(m+1)}\,\hbar \qquad (L_+)$

(2) 径向部分：

$$[E-V(r)]2mr^2+\frac{\hbar^2}{R(r)}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right)=l(l+1)\hbar^2$$

$$ER(r)=-\frac{e^2}{4\pi\epsilon_0 r}R(r)+\frac{\hbar^2 l(l+1)}{2mr^2}R(r)-\frac{\hbar^2}{2mr^2}\frac{d}{dr}\left(r^2\frac{d}{dr}R(r)\right)$$

换元: $\boxed{R(r)=\frac{u(r)}{r}}$ (专门应对球坐标径向求导).

则 $\frac{d}{dr}\left[r^2\frac{d}{dr}\left(\frac{u}{r}\right)\right]=u''r$

$$\Rightarrow\quad Eu(r)=-\frac{e^2}{4\pi\epsilon_0 r}u(r)+\frac{\hbar^2 l(l+1)}{2mr^2}u(r)-\frac{\hbar^2}{2m}u(r)$$

仅考虑 $E<0$ 的束缚态解.

def: $\boxed{\kappa=\sqrt{-2mE}/\hbar,\quad \rho=2\kappa r,\quad \lambda=\frac{1}{\kappa a_0}}$

$$\Rightarrow\quad \frac{d^2u(\rho)}{d\rho^2}=\left[\frac{1}{4}-\frac{\lambda}{\rho}+\frac{l(l+1)}{\rho^2}\right]u(\rho)$$

渐近行为: $\rho\to 0$: $u(\rho)\sim\rho^{l+1}$ $\quad\Rightarrow u(\rho)=\rho^{l+1}e^{-\rho/2}v(\rho)$

$\rho\to\infty$: $u(\rho)\sim e^{-\rho/2}$

$$\Rightarrow\quad \rho\frac{d^2v(\rho)}{d\rho^2}+[2(l+1)-\rho]\frac{dv(\rho)}{d\rho}+(\lambda-l-1)v(\rho)=0$$

合流超几何方程

(a) 级数解法.

$\boxed{v(\rho)=\sum_{j=0}^{\infty}a_j\rho^j}$ $\rightarrow$ $\frac{dv(\rho)}{d\rho}=\sum_{j=1}^{\infty}a_j\cdot j\cdot\rho^{j-1}=\sum_{j=0}^{\infty}(j+1)a_{j+1}\rho^j$

$\frac{d^2v(\rho)}{d\rho^2}=\sum_{j=1}^{\infty}j(j+1)a_{j+1}\rho^{j-1}$

$$\Rightarrow\quad j(j+1)a_{j+1}+2(l+1)(j+1)a_{j+1}-ja_j+(\lambda-l-1)a_j=0$$

$$a_{j+1}=\frac{j+l+1-\lambda}{(j+1)(j+2l+2)}a_j\ \xrightarrow{j\to\infty}\ \frac{1}{j}a_j,\quad v\to e^{\rho}$$

截断: 在某个 j 处有 $j+l+1-\lambda=0$. 称这个最大的 j 为 n_r

则 $\boxed{\lambda=n_r+l+1\triangleq n}$. $\Rightarrow n=\frac{1}{\kappa a_0}=\frac{\hbar}{\sqrt{-2mE}\,a_0}\Rightarrow E_n=-\frac{\hbar^2}{2ma_0^2}\cdot\frac{1}{n^2}$

给定 n 后, l 可取 0 到 $n-1$ (n_r 也是)

5. $\frac{d}{dt}\langle\hat{O}\rangle=\frac{1}{i\hbar}\langle[\hat{O},\hat{H}]\rangle+\langle\frac{\partial\hat{O}}{\partial t}\rangle$ （代入sch Equ即可证明）

$\psi_{nlm}=R_{nl}(r)Y_l^m(\theta,\phi)$

$n,\ l\in[0,n-1]$,
$m\in[-l,l]$.

$\boxed{R_{10}=\frac{2}{a_0^{3/2}}e^{-r/a}}$ $\quad Y_0^0=\frac{1}{\sqrt{4\pi}}$

①同个l, $2l+1$重简并，体现在 $[L^2,L_\pm]=0$而$[L_z,L_\pm]\neq0$.
即 $L^2(L_\pm Y)=c(L_\pm Y)$.
$L_z(L_\pm Y)=(m\pm1)\hbar(L_\pm Y)$

$dV=r^2\sin\theta\, dr\, d\theta\, d\varphi$

$\int_0^\infty t^{p-1}e^{-t}dt=\Gamma(p)=(p-1)!$

$\int_{-\infty}^{\infty}e^{-ax^2}dx=\sqrt{\frac{\pi}{a}}$

②同个n, n重简并，体现在 $[A_i,H]=0$ 而$[A_i,L^2]\neq0$.

$R_{nl}(\rho)\propto\rho^l e^{-\rho}\frac{1}{an}L_{n-l-1}^{2l+1}(2\rho)\quad(\rho=kr)$

$\uparrow$

$n-l-1=n_r$阶多项式
有n_r个节点.

3. 各种对易子.
现有物理量：$\hat{x},\hat{p},\hat{H},\hat{L}^2,\hat{L}_i,\hat{L}_\pm$（新增）

(1) $[\hat{L}_i,\hat{L}_j]=i\hbar\hat{L}_k\qquad[L_i,x_j]=i\hbar x_k,\quad[L_i,p_j]=i\hbar p_k$

$[\hat{L}_i,\hat{L}^2]=0$

$[L_i,\hat{p}^2]=0\qquad[\hat{L}_i,\hat{r}^2]=0.$

$[L_i,\hat{H}]=[L_i,V(\vec{r})]\ \begin{cases}V(\vec{r})=V(r):\ =0\\ V(\vec{r})=V(r,\theta):\begin{cases}=0, & i=z\\ \neq0 & i\neq z\end{cases}\end{cases}$

$[L^2,H]=[L_x^2+L_y^2+L_z^2,H]=0.$（当$V=V(r)$）.

(2) $[L_+,L_-]=2\hbar L_z$ $\qquad$ 若$[A,B]=0$，

$[L_\pm,L^2]=0$ $\qquad$ 则$[A,f(B)]=0$

$[L_z,L_\pm]=\pm\hbar L_\pm$

4. 证明算符厄米性：在某表象下，证$\int(\hat{O}f)^*g=\int f^*\hat{O}g$，常用IBP.

角动量

这一页有两个Typo，J、S的对易应该有个hbar

1. 角动量理论

$[J_i, J_j] = i\epsilon_{ijk}J_k$

$a_\pm$是H的升降算符
$J_\pm$是J_z的升降算符

惯例设 $J_z|m\rangle = m|m\rangle$

定义 $J^2 = J_x^2+J_y^2+J_z^2$　　$[J^2, J_i]=0$，故 J^2与J_i有共同本征态

$J_\pm = J_x \pm iJ_y$　　$[J_z, J_\pm] = \pm J_\pm$，$[J^2, J_\pm]=0$

则 $J_z(J_\pm|m\rangle) = (m\pm1)(J_\pm|m\rangle)$

$J_\pm|m\rangle = \sqrt{j(j+1)-m(m\pm1)}\,|m\pm1\rangle$

$J^2|j,m\rangle = j(j+1)|j,m\rangle$　　$(J_+|m=j\rangle=0)$

j固定时，$m=-j,\cdots,j$　$(2j+1个)$.

2. 自旋

$[S_i, S_j] = i\epsilon_{ijk}S_k$.　　$S^2|s,m\rangle = s(s+1)|s,m\rangle$.　$s=\frac{1}{2}$.　$m=\pm\frac{1}{2}$.

以S_z的本征矢为基，有　$S_+|\uparrow\rangle=0$, $S_-|\downarrow\rangle=0$　$S_\pm = S_x \pm iS_y$ $\Rightarrow$

$S_z = \frac{1}{2}\sigma_z = \frac{1}{2}\begin{pmatrix}1&0\\0&-1\end{pmatrix}$　$S_x = \frac{1}{2}\sigma_x = \frac{1}{2}\begin{pmatrix}0&1\\1&0\end{pmatrix}$　$S_y = \frac{1}{2}\sigma_y = \frac{1}{2}\begin{pmatrix}0&-i\\i&0\end{pmatrix}$

本征矢 $\begin{pmatrix}1\\0\end{pmatrix}$, $\begin{pmatrix}0\\1\end{pmatrix}$　　本征矢 $\frac{1}{\sqrt{2}}(1,1)$.
$\frac{1}{\sqrt{2}}(1,-1)$.

$$\begin{cases}\sigma_i\sigma_j = i\epsilon_{ijk}\sigma_k + I\delta_{ij}\\ [\sigma_i, \sigma_j] = 2i\epsilon_{ijk}\sigma_k\\ (\sigma\cdot n)^2 = I\end{cases}$$

$(\vec{\sigma}\cdot\vec{A})(\vec{\sigma}\cdot\vec{B}) = I(\vec{A}\cdot\vec{B}) + i\vec{\sigma}(\vec{A}\times\vec{B})$
$([\sigma, A]=0,\ [\sigma, B]=0)$

$\sigma\cdot n = \begin{pmatrix}\cos\theta & \sin\theta e^{-i\phi}\\ \sin\theta e^{i\phi} & -\cos\theta\end{pmatrix}$.　本征态：$\lambda_1=1$. $|+\rangle_n = \begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}e^{i\phi}\end{pmatrix}$

$\lambda_2=-1$. $|-\rangle_n = \begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}e^{i\phi}\end{pmatrix}$.

S_n本征值总是$\frac{1}{2}$, $-\frac{1}{2}$. 只是概率不同

$P(某算符的某本征值_i) = |\langle 算符本征态_i|现在态\rangle|^2$

$\langle S_n\rangle = \frac{1}{2}\langle\psi|\sigma_n|\psi\rangle = P(\frac{1}{2})\cdot\frac{1}{2} + P(-\frac{1}{2})\cdot(-\frac{1}{2})$

$\frac{1}{2}\,{}_z\langle\uparrow|\sigma_n|\uparrow\rangle_z = \frac{1}{2}\cos\theta$

$$(A\otimes B)(C\otimes D)=AC\otimes BD.$$

3. 两个角动量耦合

$$A\otimes B=\begin{pmatrix}A_{11} & A_{12}\\ A_{21} & A_{22}\end{pmatrix}\otimes B=\begin{pmatrix}A_{11}B & A_{12}B\\ A_{21}B & A_{22}B\end{pmatrix}$$

空间 H_1 H_2 H

角动量 J_{1i} J_{2i} $J_{1i}\otimes I+I\otimes J_{2i}$

基 $|j_1m_1\rangle\otimes|j_2m_2\rangle$. $|j_1m_1j_2m_2\rangle$ $\{J_1^2,J_2^2,J_{1z},J_{2z}\}$

在H中, 也有 $[J_i,J_j]=i\epsilon_{ijk}J_k$.

$$J^2=J_x^2+J_y^2+J_z^2=J_1^2\otimes I+I\otimes J_2^2+\sum_i 2J_{1i}\otimes J_{2i}$$ 不可交换

耦合基: $|j_1m_1j_2m_2\rangle$. $\{J_1^2,J_{1z},J_2^2,J_{2z}\}$

eg. $\vec{L}$ 与 $\vec{S}$. $|l s j m\rangle$. $\{L^2,S^2,J^2,J_z\}$.

$$2\vec{S}\otimes\vec{L}=\vec{\sigma}\otimes\vec{L}=\sigma_x\otimes L_x+\sigma_y\otimes L_y+\sigma_z\otimes L_z$$
$$=\begin{pmatrix}L_z & L_x-iL_y\\ L_x+iL_y & -L_z\end{pmatrix}=\begin{pmatrix}L_z & L_-\\ L_+ & -L_z\end{pmatrix}$$

$$(\vec{\sigma}\otimes\vec{L})^2=\begin{pmatrix}L_z & L_-\\ L_+ & -L_z\end{pmatrix}\begin{pmatrix}L_z & L_-\\ L_+ & -L_z\end{pmatrix}=\begin{pmatrix}L_z^2-L_-L_+ & L_zL_--L_-L_z\\ L_+L_z-L_zL_+ & L_+L_-+L_z^2\end{pmatrix}$$

$[L_z,L_\pm]=\pm L_\pm$

$$L^2=L_x^2+L_y^2+L_z^2=L_z^2+L_+L_--L_z$$
$$=L_z^2+L_-L_++L_z$$

$$=I\otimes L^2-\vec{\sigma}\otimes\vec{L}$$

$$\Rightarrow x^2=l(l+1)-x\Rightarrow x=\begin{cases}l\\-(l+1)\end{cases}$$

$$\Rightarrow J^2=S^2\otimes I+I\otimes L^2+\vec{\sigma}\otimes\vec{L}$$

$$j(j+1)=s(s+1)+l(l+1)+x$$

$x=l$: $j(j+1)=\frac{3}{4}+l(l+2)=(l+\frac{3}{2})(l+\frac{1}{2})\Rightarrow j=l+\frac{1}{2}$

$x=-l-1$: $j(j+1)=\frac{3}{4}+(l-1)(l+1)=(l+\frac{1}{2})(l-\frac{1}{2})\Rightarrow j=l-\frac{1}{2}$.

- 具体如何找 $|j_1j_2jm\rangle$ 与 $|j_1m_1j_2m_2\rangle$ 基之间的关系?

j_1,j_2 是给定的. $J_z|m_1\rangle|m_2\rangle=(m_1+m_2)|m_1\rangle|m_2\rangle$, $J_z|j,m\rangle=m|j,m\rangle$

1) max: $J_z|j_1\rangle|j_2\rangle=(j_1+j_2)|j_1\rangle|j_2\rangle$, $J_z|\bar{j},\bar{j}\rangle=\bar{j}|\bar{j},\bar{j}\rangle\Rightarrow\bar{j}=j_1+j_2$

$|\bar{j},\bar{j}\rangle=|j_1\rangle|j_2\rangle$

2) $\bar{j}$ 的其他耦合基:

$$J_-|\bar{j},\bar{j}\rangle=\sqrt{2\bar{j}}\,|\bar{j},\bar{j}-1\rangle=(J_{1-}+J_{2-})|j_1\rangle|j_2\rangle=\sqrt{2j_1}\,|j_1-1\rangle|j_2\rangle+\sqrt{2j_2}\,|j_1\rangle|j_2-1\rangle$$

$|\bar{j},\bar{j}-1\rangle=\sqrt{\frac{j_1}{\bar{j}}}|j_1-1\rangle|j_2\rangle+\sqrt{\frac{j_2}{\bar{j}}}|j_1\rangle|j_2-1\rangle$. $\bar{j}=j_1+j_2$

继续作用 J_-，直到 $|\bar{j},-\bar{j}\rangle=|-j_1\rangle|-j_2\rangle$

3) 找 $j=\bar{j}-1$ 的梯子.

顶端: $|\bar{j}-1,\bar{j}-1\rangle$.

$J_z|\bar{j}-1,\bar{j}-1\rangle=(\bar{j}-1)|\bar{j}-1,\bar{j}-1\rangle=(j_1+j_2-1)|\bar{j}-1,\bar{j}-1\rangle$

故 $|\bar{j}-1,\bar{j}-1\rangle$ 为 $|j_1-1\rangle|j_2\rangle$ 与 $|j_1\rangle|j_2-1\rangle$ 线性组合

其 j 与 $|\bar{j},\bar{j}-1\rangle$ 不同. 故与其正交. 二维子空间 (两个基组成).

故找一个正交的是唯一的. $|\bar{j}-1,\bar{j}-1\rangle=\sqrt{\frac{j_2}{\bar{j}}}|j_1-1\rangle|j_2\rangle-\sqrt{\frac{j_1}{\bar{j}}}|j_1\rangle|j_2-1\rangle$

$(a\vec{e}_1+b\vec{e}_2)\cdot(c\vec{e}_1+d\vec{e}_2)=0\Rightarrow ac+bd=0$. 取 $c=b$, $d=-a$ 即可.

继续作用 J_-. 直到 $|\bar{j}-1,-\bar{j}+1\rangle$

4) 继续找其他梯子.

$|\bar{j}-2,\bar{j}-2\rangle$ 与 $|\bar{j},\bar{j}-2\rangle$, $|\bar{j}-1,\bar{j}-2\rangle$ 均正交. 已用2个 3维 $\begin{cases} j_1,\ j_2-2 \\ j_1-1,\ j_2-1 \\ j_1-2,\ j_2 \end{cases}$

$|\bar{j}-3,\bar{j}-3\rangle$ 与 $|\bar{j},\bar{j}-3\rangle$, $|\bar{j}-1,\bar{j}-3\rangle$, $|\bar{j}-2,\bar{j}-3\rangle$ 均正交. 4维. $j_1,\cdots,j_1-3$
已用3个

⋮

不妨设 $j_1>j_2$.

min: $|\underline{j},\underline{j}\rangle$. $\underline{j}=|j_1-j_2|=j_1+j_2-2j_2$. $(2j_2+1)$维. $j_1,\cdots,j_1-2j_2$
已用 $2j_2$ 个. $-j_2,\cdots,j_2$

如果再小. 如 $|j_1-j_2-1,j_1-j_2-1\rangle$. $(2j_2+1)$维 $j_1-1,\cdots,j_1-2j_2-1$
已用 $(2j_2+1)$ 个. $-j_2,\cdots,j_2$

无法再构造正交基了. 故 $\underline{j}=|j_1-j_2|$

总基矢数 $\begin{cases} \text{耦合基}: \sum\limits_{j=|j_1-j_2|}^{j_1+j_2}(2j+1)=(2j_2+1)\cdot(j_1+\cancel{j_2}+j_1-\cancel{j_2})+(2j_2+1)=(2j_1+1)\cdot(2j_2+1) \\ \text{非耦合基}: \begin{matrix} m_1: 2j_1+1 \\ m_2: 2j_2+1 \end{matrix} \Rightarrow (2j_1+1)(2j_2+1)\text{ 个} \end{cases}$ //

数学图像：eg. $j_1=2$ $j_2=\frac{3}{2}$

m_2 \ m_1	2	1	0	-1	-2
$\frac{3}{2}$	$\frac{7}{2}$	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$
$\frac{1}{2}$	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$
$-\frac{1}{2}$	$\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$
$-\frac{3}{2}$	$\frac{1}{2}$	$-\frac{1}{2}$	$-\frac{3}{2}$	$-\frac{5}{2}$	$-\frac{7}{2}$

非耦合基.

每个m_1的子空间一样大(m_2维)

每个m_2的子空间一样大(m_1维)

m \ j	$\frac{7}{2}$	$\frac{5}{2}$	$\frac{3}{2}$	$\frac{1}{2}$
$\frac{7}{2}$	○			
$\frac{5}{2}$	□	□		
$\frac{3}{2}$	△	△	△	
$\frac{1}{2}$	√	√	√	√
$-\frac{1}{2}$	√	√	√	√
$-\frac{3}{2}$	△	△	△	
$-\frac{5}{2}$	□	□		
$-\frac{7}{2}$	○			

耦合基

每个j的子空间不一样大($2j+1$维)

每个m的子空间不一样大

($j_1+j_2-|m|+1$维)

同种符号重新线性组合

只在m同的子空间内重新组合，

不会混合m不同的子空间.

eg. l，$s=\frac{1}{2}$ 的CG系数.

$j=l\pm\frac{1}{2}$，$m_j=m_l+m_s$，$|j\,m_j\rangle=\sum_{m_s}|m_l m_s\rangle\langle m_l m_s|j\,m_j\rangle$

$m_l=m_j-m_s$

求它，j与m_j当已知

$$|j,m_j\rangle=\underbrace{|m_j-\tfrac{1}{2},+\tfrac{1}{2}\rangle}_{|e_1\rangle}\underbrace{\langle m_j-\tfrac{1}{2},+\tfrac{1}{2}|j\,m_j\rangle}_{C_1}$$

$$+\underbrace{|m_j+\tfrac{1}{2},-\tfrac{1}{2}\rangle}_{|e_2\rangle}\underbrace{\langle m_j+\tfrac{1}{2},-\tfrac{1}{2}|j\,m_j\rangle}_{C_2}=\begin{pmatrix}C_1\\C_2\end{pmatrix}$$

$J^2|j,m_j\rangle=j(j+1)|j,m_j\rangle \Rightarrow$ 得一矩阵方程. 解出$C_1(j,m_j)$, $C_2(j,m_j)$

($J_{11}=\langle e_1|J|e_1\rangle,\cdots$)

$\Rightarrow\begin{cases}j=l+\frac{1}{2}\text{时},\ C_1=\quad,\ C_2= \\ j=l-\frac{1}{2}\text{时},\ C_1=\quad,\ C_2=\end{cases}$ (带m_j)

$\langle j\,m_j|S_z|j\,m_j\rangle=\pm\frac{m_j}{2l+1}$ ($j=l\pm\frac{1}{2}$).

$A|a_i\rangle = a_i|a_i\rangle$
$B|b_i\rangle = b_i|b_i\rangle$

$$(A\otimes I + I\otimes B)(|a\rangle\otimes|b\rangle)$$
$$= (A\otimes I)(|a\rangle\otimes|b\rangle) + (I\otimes B)(|a\rangle\otimes|b\rangle)$$
$$= a|a\rangle|b\rangle + b|a\rangle|b\rangle$$
$$= (a+b)|a\rangle|b\rangle$$

故 $A\otimes I + I\otimes B$ 的本征矢是 $|a_i\rangle\otimes|b_j\rangle$，对应本征值 (a_i+b_j).

eg. 自旋单态为什么是纠缠态？

$|\psi\rangle = \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_z - |\downarrow\uparrow\rangle_z)$　　沿任意方向测自旋，总得一上一下.

$= \frac{1}{\sqrt{2}}(0,1,-1,0)^T$.

$S_n = S_{1n}\otimes I + I\otimes S_{2n}$

S_{in}本征矢：$i=1,2$　$\frac{1}{2}$：$\begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}e^{i\phi}\end{pmatrix}$　　$-\frac{1}{2}$：$\begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}e^{i\phi}\end{pmatrix}$

4个本征矢：$|\uparrow\uparrow\rangle_n$, $|\uparrow\downarrow\rangle_n$, $|\downarrow\uparrow\rangle_n$, $|\downarrow\downarrow\rangle_n$.

在S_z表象下的分量：

$$|\uparrow\downarrow\rangle_n = \begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}e^{i\phi}\end{pmatrix}\otimes\begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}e^{i\phi}\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\sin\theta\\ -\cos^2\frac{\theta}{2}e^{i\phi}\\ \sin^2\frac{\theta}{2}e^{i\phi}\\ -\frac{1}{2}\sin\theta e^{2i\phi}\end{pmatrix}$$

$$|\downarrow\uparrow\rangle_n = \begin{pmatrix}\sin\frac{\theta}{2}\\ -\cos\frac{\theta}{2}e^{i\phi}\end{pmatrix}\otimes\begin{pmatrix}\cos\frac{\theta}{2}\\ \sin\frac{\theta}{2}e^{i\phi}\end{pmatrix} = \begin{pmatrix}\frac{1}{2}\sin\theta\\ \sin^2\frac{\theta}{2}e^{i\phi}\\ -\cos^2\frac{\theta}{2}e^{i\phi}\\ -\frac{1}{2}\sin\theta e^{2i\phi}\end{pmatrix}$$

则 $\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle_n - |\downarrow\uparrow\rangle_n) = \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix}e^{i\phi} \simeq \frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\\0\end{pmatrix}$.　等价的.

故 $P(\frac{1}{2},-\frac{1}{2}) = |{}_n\langle\uparrow\downarrow|\psi\rangle|^2 = |{}_z\langle\uparrow\downarrow|\psi\rangle|^2 = \frac{1}{2}$.

$P(-\frac{1}{2},\frac{1}{2}) = |{}_n\langle\downarrow\uparrow|\psi\rangle|^2 = |{}_z\langle\downarrow\uparrow|\psi\rangle|^2 = \frac{1}{2}$.

但三重态中的态 $(|\uparrow\downarrow\rangle+|\uparrow\downarrow\rangle)$ 则并不是在任何方向测都一上一下.

$$P(\tfrac{1}{2},-\tfrac{1}{2})=\left|{}_n\langle\uparrow\downarrow|\psi\rangle\right|^2$$

$$=\left|\left(\tfrac{1}{2}\sin\theta,-\cos^2\tfrac{\theta}{2}e^{-i\phi},\sin^2\tfrac{\theta}{2}e^{-i\phi},-\tfrac{1}{2}\sin\theta e^{-2i\phi}\right)\begin{pmatrix}0\\1\\1\\0\end{pmatrix}\right|^2\cdot\frac{1}{2}$$

$$=\left|-\cos^2\tfrac{\theta}{2}e^{-i\phi}+\sin^2\tfrac{\theta}{2}e^{-i\phi}\right|^2\cdot\frac{1}{2}$$

$$=\frac{1}{2}\left|\cos\theta e^{-i\phi}\right|^2=\frac{1}{2}\cos^2\theta.$$ 不一定是$\frac{1}{2}$

$$P(-\tfrac{1}{2},\tfrac{1}{2})=\left|{}_n\langle\downarrow\uparrow|\psi\rangle\right|^2$$

$$=\left|\left(\tfrac{1}{2}\sin\theta,\sin^2\tfrac{\theta}{2}e^{-i\phi},-\cos^2\tfrac{\theta}{2}e^{-i\phi},-\tfrac{1}{2}\sin\theta e^{-2i\phi}\right)\begin{pmatrix}0\\1\\1\\0\end{pmatrix}\right|^2\cdot\frac{1}{2}$$

$$=\frac{1}{2}\cdot\left|-\cos\theta e^{-i\phi}\right|^2=\frac{1}{2}\cos^2\theta$$

$P_1+P_2=\cos^2\theta\neq1$. 只有 $\theta=0$ 时 ($S_n=S_z$时) 才必反向.

如果 $\theta=\frac{\pi}{2}$. 则必同向. 而不是反向.

其他方向, 不算严格的纠缠态.

近似方法

一. 微扰论.

(一) 定态非简并微扰

$$\begin{cases} H|\psi_n\rangle = E_n|\psi_n\rangle \\ H = H_0 + W \\ H_0|n\rangle = E_n^{(0)}|n\rangle \\ E_n = E_n^{(0)} + E_n^{(1)} + \cdots \\ |\psi_n\rangle = |n\rangle + |\psi_n^{(1)}\rangle + \cdots \end{cases} \Rightarrow \begin{cases} E_n^{(1)} = \langle n|W|n\rangle \\ |\psi_n^{(1)}\rangle = \sum_{m\neq n} C_m|m\rangle \\ \qquad C_m = \dfrac{\langle m|W|n\rangle}{E_n^{(0)} - E_m^{(0)}} \\ E_n^{(2)} = \sum_{m\neq n} \dfrac{|\langle m|W|n\rangle|^2}{E_n^{(0)} - E_m^{(0)}} \end{cases}$$

能级越近影响越大

(二) 定态简并微扰

$$\begin{cases} H|\psi_{n,i}\rangle = E_{n,i}|\psi_{n,i}\rangle \\ H = H_0 + W \\ H_0|\phi_{n,i}\rangle = \varepsilon_n|\phi_{n,i}\rangle \\ E_{n,i} = E_{n,i}^{(0)} + E_{n,i}^{(1)} + \cdots \\ |\psi_{n,i}\rangle = |\psi_{n,i}^{(0)}\rangle + |\psi_n^{(1)}\rangle + \cdots \end{cases}$$

$\Rightarrow E_{n,i}^{(1)}$: $\sum_k \langle\phi_{n,j}|W|\phi_{n,k}\rangle\langle\phi_{n,k}|\psi_{n,i}^{(0)}\rangle = E_{n,i}^{(1)}\langle\phi_{n,j}|\psi_{n,i}^{(0)}\rangle$.

$E_{n,i}^{(1)}$是W在ε_n的简并子空间中矩阵本征值.

$i=1,2,\cdots,s$ (ε_n的简并度)

取$|\psi_{n,i}^{(0)}\rangle$时, W在ε_n子空间内对角.

ε_n子空间内: $W|\psi_{n,i}^{(0)}\rangle = E_{n,i}^{(1)}|\psi_{n,i}^{(0)}\rangle$

$$E_{n,i}^{(2)} = \sum_{m\neq n}\sum_j \frac{|W_{ni,mj}|^2}{E_n^{(0)} - E_m^{(0)}}$$

tips: 找A.使 $\begin{cases} [A,H_0]=0 \\ [A,W]=0 \\ A|\phi_{n,i}\rangle = a_{n,i}|\phi_{n,i}\rangle \end{cases}$ (CSCO)

且$a_{n,i}\neq a_{n,j}$ ($i\neq j$)

则$\langle\phi_{n,j}|W|\phi_{n,k}\rangle$就是对角的

$|\psi_{n,i}^{(0)}\rangle = |\phi_{n,i}\rangle$.

(即$j\neq k$时$W_{jk}=0$)

eg1. 塞曼效应. $\underbrace{\text{氢原子}}_{H_0}$在强B中.

↳可忽略电子自旋. $W = \dfrac{eB}{2m_e}L_z$

同个$E_n^{(0)}$.简并. CSCO一般用$\{\underset{n}{H_0}, \underset{l}{L^2}, \underset{m_l}{L_z}\}$. 均与W对易. 且完全分辨各态.

故$E_{n,i}^{(1)} = \langle n,l,m_l|W|n,l,m_l\rangle = \dfrac{eB}{2m_e}m_l$,

$n=2$ $l=0$

$l=1$. $m_l=-1,0,1$. 3条

eg2. 斯塔克效应. 氢原子在外电场中. $W=eEz$

不考虑 e^- 自旋. CSCO 也用 $\{H_0, L^2, L_z\}$. $|n, l, m_l\rangle$.

但 $[H_0, z]\neq 0$, $[L^2, z]\neq 0$. 只有 $[L_z, z]=0$. 故不同 m_l 的矩阵元为 0.

L_z 破除了一些简并. 但还有不同 l 的情况. 没有升降算符用. 积分算矩阵元.

eg. 对 $n=2$ 能级: $\begin{cases} l=0, & m=0 \\ l=1, & m=-1,0,1 \end{cases}$ 即 $\begin{cases} m=0. & l=0, l=1 \\ m=-1, & l=1 \\ m=1, & l=1 \end{cases}$

$n=2, m_l=0$ 子空间: $\langle 0|W|0\rangle = 0$

$\langle 0|W|1\rangle \propto \int r\cos\theta (1-\frac{r}{2a_0})e^{-r/a_0}\ r\sin\theta e^{i\phi}\ z\ r\sin\theta\ dr d\theta d\phi$
$= -3a_0 eE$

$\langle 1|W|0\rangle = \cdots = -3eEa_0$

$\langle 1|W|1\rangle = 0.$

$m_l=-1$: $l=1$. $E^{(1)}_{2,1,-1} = \langle 21-1|W|21-1\rangle = 0$

$m_l=1$: $l=1$. $E^{(1)}_{2,1,1} = \langle 211|W|211\rangle = 0$.

利用好奇偶性. 先将 0 的写出来. 大部分是 0.

eg3. 自旋轨道耦合修正. $W_{so}\propto \frac{1}{r^3}\vec{S}\cdot\vec{L}$

eg4. 反常塞曼效应: 氢原子在弱 $\vec{B}$ (比 $\vec{S}\cdot\vec{L}$ 耦合还弱)

氢原子奇偶性: $\vec{r}\to -\vec{r}$: $(r,\theta,\phi)\to(r,\pi-\theta,\pi+\phi)$

$$Y_l^m(\pi-\theta,\pi+\phi) = (-1)^l Y_l^m(\theta,\phi)$$

因为 $Y_l^m \sim P_l^m(\cos\theta)e^{im\phi}$. $e^{im(\phi+\pi)}=(-1)^m e^{im\phi}$, $P_l^m(\cos(\pi-\theta))=P_l^m(-\cos\theta)$

$P_l^m(-x) = (-1)^{l+m}P_l^m(x)$

二. 变分法.

用于估算基态能量. $\langle\psi|H|\psi\rangle \geqslant E_0$, $\forall\psi\in\mathcal{H}$

$$\Rightarrow \langle H\rangle_\lambda = \langle\psi(\lambda)|H|\psi(\lambda)\rangle.$$

$$\left.\frac{\partial\langle H\rangle_\lambda}{\partial\lambda}\right|_{\lambda_0}=0 \Rightarrow \langle H\rangle_{\lambda_0}\geqslant E_0$$

若选 $|\psi\rangle$ 为 H_0 的 $|n\rangle$. 则得的即微扰论一阶修正

三. WKB近似.

$$\phi''(x)+f(x)\phi(x)=0$$

$$\Rightarrow \phi(x)\sim f^{-\frac{1}{4}}(x)e^{\pm i\int dx\sqrt{f(x)}}$$

(二) 含时微扰

$$H=H_0+W(t)$$

$$H_0|n\rangle=E_n^{(0)}|n\rangle$$ 则 $|\psi_0(t)\rangle=\sum_n e^{-iE_n^{(0)}t}a_n|n\rangle$

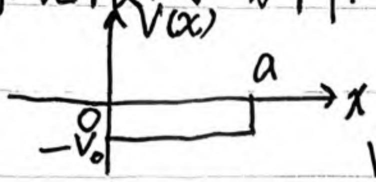
ansats: $|\psi(t)\rangle=\sum_n e^{-iE_n^{(0)}t}a_n(t)|n\rangle$.

代回 $(H_0+W)|\psi\rangle=i\partial_t|\psi\rangle$.

$$\Rightarrow a_s'(t)=\sum_n -i e^{-i(E_n^{(0)}-E_s^{(0)})t}a_n(t)W_{sn}$$ (严格的)

进行一些近似.

eg. "无限浅"势阱.



求基态能级

$V_0\to 0$

法一：严格解展开.

本征值离散化是边条导致的. $\begin{cases}\xi\tan\xi=\eta\\ \xi^2+\eta^2=\frac{ma^2V_0}{2\hbar^2}\end{cases}$ $V_0\to 0$ 时只有一个交点. 基态.

$\tan\xi\approx\xi$

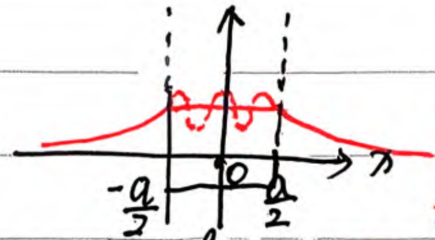
$\Rightarrow \xi^2=\eta$.

$\xi^4=\eta^2=\frac{ma^2V_0}{2\hbar^2}-\xi^2$

$\Rightarrow \xi^2=\frac{-1\pm\sqrt{1+2ma^2V_0/\hbar^2}}{2}$ $\xi^2>0$ 取正根. $\frac{a^2}{4}\cdot\frac{2m(E+V_0)}{\hbar^2}=\frac{-1+\sqrt{1+2ma^2V_0^2/\hbar^2}}{2}$

展开: $\frac{ma^2}{\hbar^2}(E+\cancel{V_0})\approx\cancel{-1}+\cancel{1}+\cancel{\frac{ma^2V_0^2}{\hbar^2}}-\frac{m^2a^4V_0^2}{2\hbar^4}$

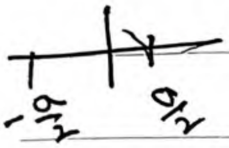
$$E\approx-\frac{ma^2}{2\hbar^2}V_0^2$$



$k=\frac{\sqrt{2mE}}{\hbar}$

注意0选中间

法二: $\psi(x)=\begin{cases}e^{+kx}, & x<-\frac{a}{2}\\ e^{-kx}, & x>\frac{a}{2}\\ \text{振荡}, & -\frac{a}{2}<x<\frac{a}{2}\end{cases}$ → 用常数 $e^{-k\frac{a}{2}}$ 近似 基态无节点.



$$\begin{cases}\text{则 } \psi'(x)\Big|_{-\frac{a}{2}}^{a/2}=-ke^{-kx}\Big|_{\frac{a}{2}}-(+k)e^{+kx}\Big|_{-\frac{a}{2}}=-ke^{-k\frac{a}{2}}-ke^{-k\frac{a}{2}}=-2ke^{-\frac{ka}{2}}\\ \text{又有: } \int_{-\frac{a}{2}}^{\frac{a}{2}}\psi''(x)+\frac{2m(E-V)}{\hbar^2}\psi(x)=0 \qquad (E\ll V)\end{cases}$$

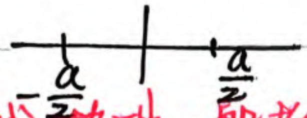
$$\psi'(x)\Big|_{-\frac{a}{2}}^{\frac{a}{2}}\approx\frac{2m}{\hbar^2}\int_{-\frac{a}{2}}^{\frac{a}{2}}dx\,V(x)\cancel{e^{-k\frac{a}{2}}}=-2k\cancel{e^{-\frac{ka}{2}}}$$

$$\Rightarrow \frac{\sqrt{2mE}}{\hbar}=-\frac{m}{\hbar^2}\int dx\,V(x)$$

$$\boxed{E_g=-\frac{m}{2\hbar^2}\left(\int dx\,V(x)\right)^2}=-\frac{m}{2\hbar^2}\cdot V_0^2a^2$$

无限浅势阱基态能量均可用(不方也行)

法三：变分法.



基态：偶宇称、无节点、 无限浅势阱一般都可以用 就这里吧啊.

$\psi(x) = \sqrt{\kappa}\, e^{-\kappa|x|}$ （在$x=0$处导数不连续，是δ势阱的基态波函数）

$$\langle H\rangle_\kappa = \langle T\rangle + \langle V\rangle = \frac{\hbar^2}{2m}\int |\psi'(x)|^2 dx + 2\int_0^{\frac{a}{2}} -V_0\cdot\kappa\cdot e^{-2\kappa x}$$

$$\psi'(x) = \begin{cases} -\kappa^{3/2} e^{-\kappa x}, & x>0 \\ \kappa^{3/2} e^{\kappa x}, & x<0 \end{cases}$$

$$|\psi'(x)|^2 = \kappa^3 e^{2\kappa|x|}$$

$$= \frac{\hbar^2}{2m}\cdot 2\int_0^\infty \kappa^3 e^{2\kappa x} dx - 2\kappa V_0 \int_0^{\frac{a}{2}} e^{-2\kappa x} dx$$

$$= \frac{\hbar^2\kappa^2}{2m} + \frac{2\kappa V_0}{2\kappa}\left(e^{-\kappa a} - 1\right)$$

$$= \frac{\hbar^2\kappa^2}{2m} + V_0\left(e^{-\kappa a} - 1\right)$$

$$\frac{\partial\langle H\rangle_\kappa}{\partial\kappa} = \frac{\hbar^2}{m}\kappa - aV_0 e^{-\kappa a} = 0 \qquad \kappa \ll 1 \Rightarrow e^{-\kappa a} \simeq 1-\kappa a$$

$$\left(\frac{\hbar^2}{m} + a^2V_0\right)\kappa = aV_0$$

$$(\hbar^2 + a^2V_0)\kappa = maV_0 \qquad \kappa = \frac{maV_0}{a^2V_0+\hbar^2} = \frac{maV_0/\hbar^2}{1+a^2V_0/\hbar^2}$$

$$\approx \frac{maV_0}{\hbar^2}\left(1 - \frac{a^2V_0}{\hbar^2}\right) \approx \frac{maV_0}{\hbar^2} - \cancel{\frac{ma^3V_0^2}{\hbar^4}}$$

$$\langle H\rangle_{\kappa_0} = \underbrace{\frac{\hbar^2}{2m}\left(\frac{maV_0}{\hbar^2}\right)^2}_{V_0^2\text{阶}} + \underbrace{V_0\left[\cancel{1} - \kappa a + \frac{(\kappa a)^2}{2}\right]}_{\text{也留到}V_0^2\text{阶·里面}V_0\text{阶的约了.}}$$

$$= \frac{\hbar^2}{2m}\cdot\frac{m^2a^2V_0^2}{\hbar^4} - V_0 a\frac{maV_0}{\hbar^2} - \cancel{V_0\cdot\frac{1}{2}a^2\cdot\frac{m^2a^2V_0^2}{\hbar^2}}$$

$$= -\frac{ma^2V_0^2}{2\hbar^2} \checkmark$$

法四：微扰论.

$$H_0 = \frac{p^2}{2m}. \qquad W = \begin{cases} -V_0, & -\frac{a}{2}<x<\frac{a}{2} \\ 0, & \text{其他.} \end{cases}$$

以前学的微扰论一般只用于束缚态（未扰动时）. 对于连续谱，$E_n^{(1)} = W_{nn}$，

$C_m = \sum \frac{\langle m|W|n\rangle}{E_n^{(0)}-E_m^{(0)}}$，分母会发散，实际应该用散射问题的框架.

全同粒子.

☆两个全同粒子构成的态 = 交换两个粒子，态只差整体相位的态 (1)

· $H_1 \otimes H_2$ 可视为两粒子所有态的直积，也可视为对称子空间与反对称子空间的直和. 但物理允许的态（全同粒子态）只能在对称或反对称子空间里，不能是两子空间矢量的叠加，否则违反 (1)

eg. $|\uparrow\downarrow\rangle$ 不可以存在. 因为它是 $\frac{1}{\sqrt{2}}\left[\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle+|\downarrow\uparrow\rangle)+\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle-|\downarrow\uparrow\rangle)\right)$

· (1) 的数学表述：全同粒子态 $|\psi\rangle$ 满足 $P|\psi\rangle = \pm|\psi\rangle$. 是置换算符的本征矢. P 在对称 & 反对称基下是 $\begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{matrix}\text{对称基} \\ \text{反对称基}\end{matrix}$.

☆自旋统计关系：

对称	自旋 $\frac{2n}{2}$	BE	$\frac{1}{e^{\beta E}-1}$
反对称	自旋 $\frac{2n+1}{2}$	FD	$\frac{1}{e^{\beta E}+1}$ $a_s < 1$

作业题

- 概念

假设两个自旋为 $\frac{1}{2}$ 的粒子处于单态构型 $|00\rangle = \frac{1}{\sqrt{2}}(|+\rangle_1|-\rangle_2 - |-\rangle_1|+\rangle_2)$. 令 $S_a^{(1)}$ 表示粒子 1 在由矢量 $\mathbf{a}$ 所定义方向上的自旋角动量分量. 类似地, 令 $S_b^{(2)}$ 表示粒子 2 在方向 $\mathbf{b}$ 上的自旋角动量分量. 证明:

$$\left\langle S_a^{(1)} S_b^{(2)} \right\rangle = -\frac{\hbar^2}{4}\cos\theta$$

其中 θ 是矢量 $\mathbf{a}$ 与 $\mathbf{b}$ 之间的夹角.

3 周期性边界条件下的简并微扰

有一个质量为 m 的粒子在 $-\frac{L}{2} \le x \le \frac{L}{2}$ 自由运动, 波函数满足周期性边界条件:

$$\psi(-\frac{L}{2}) = \psi(\frac{L}{2})$$

现在加入一个微扰势场, 表达式为:$V(x) = -V_0 e^{-x^2/a^2} \quad (a \ll \frac{L}{2})$, 计算所有**激发态**的能量一阶修正. 提示: 由于微扰快速衰减, 计算矩阵元做积分时可以将积分上下限推广至无穷.

- 适合练计算

2. A particle in the infinite square well has the initial wave function

$$\Psi(x,0) = \begin{cases} Ax, & 0 \le x \le a/2, \\ A(a-x), & a/2 \le x \le a. \end{cases}$$

(a) Sketch $\Psi(x,0)$, and determine the constant A.

(b) Find $\Psi(x,t)$.

(c) Find the expectation value of the energy $\langle H \rangle$

1 角动量耦合

计算 $j_1 = 1$ 和 $j_2 = \frac{1}{2}$ 耦合的 CG 系数.

2 变分法 2

采用变分法的思想估算氦原子的基态能级 (忽略原子核的运动), 并与真实值 $-78.6\ eV$ 比较. 提示: 在选取试探波函数时, 可以把体系近似作为两个独立的类氢原子体系, 再把合适的参数做变分求得.

- 小坑

6. Consider a state which is given in terms of three orthonormal vectors $|\phi_1\rangle$, $|\phi_2\rangle$, and $|\phi_3\rangle$ as follows:

$$|\psi\rangle = \frac{1}{\sqrt{15}}|\phi_1\rangle + \frac{1}{\sqrt{3}}|\phi_2\rangle + \frac{1}{\sqrt{5}}|\phi_3\rangle,$$

where $|\phi_n\rangle$ are eigenstates to an operator $\hat{B}$ such that: $\hat{B}|\phi_n\rangle = (3n^2 - 1)|\phi_n\rangle$ with $n = 1, 2, 3$.

(a) Find the norm of the state $|\psi\rangle$.
(b) Find the expectation value of $\hat{B}$ for the state $|\psi\rangle$.
(c) Find the expectation value of $\hat{B}^2$ for the state $|\psi\rangle$.

3. The Hamiltonian for a certain three-level system is represented by the matrix

$$\mathbf{H} = \hbar\omega \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

Two other observables, A and B, are represented by the matrices

$$\mathbf{A} = \lambda \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 2 \end{pmatrix}, \quad \mathbf{B} = \mu \begin{pmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix},$$

where ω, λ, and μ are positive real numbers.

(a) Find the eigenvalues and (normalized) eigenvectors of **H**, **A**, and **B**.

(b) Suppose the system starts out in the generic state

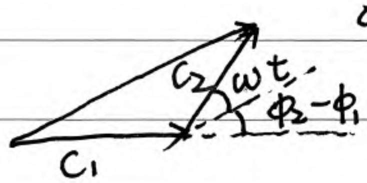
$$|\mathcal{S}(0)\rangle = \begin{pmatrix} c_1 \\ c_2 \\ c_3 \end{pmatrix},$$

with $|c_1|^2 + |c_2|^2 + |c_3|^2 = 1$. Find the expectation values (at $t = 0$) of H, A, and B.

(c) What is $|\mathcal{S}(t)\rangle$? If you measured the energy of this state (at time t), what values might you get, and what is the probability of each? Answer the same questions for A and for B.

② A：可能值：本征值 $\lambda, -\lambda, 2\lambda$

可能性：$P(\lambda) = |\langle\psi_\lambda|s(t)\rangle|^2 = \left|\frac{1}{\sqrt{2}}(1,1,0)\begin{pmatrix}c_1e^{-i\omega t}\\ c_2e^{-i2\omega t}\\ c_3e^{-i2\omega t}\end{pmatrix}\right|^2$



$\phi_2-\phi_1$：c_2和c_1的辐角差

$$= \frac{1}{2} \times |c_1e^{-i\omega t} + c_2e^{-i2\omega t}|^2$$

$$= \frac{1}{2}\left(|c_1|^2+|c_2|^2+2|c_1c_2|\cos(\phi_2-\phi_1+\omega t)\right)$$